Università degli studi di Padova

Stochastic Methods for Finance

Report 5: Option Greeks and Volatility surface analysis

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1 Introduction

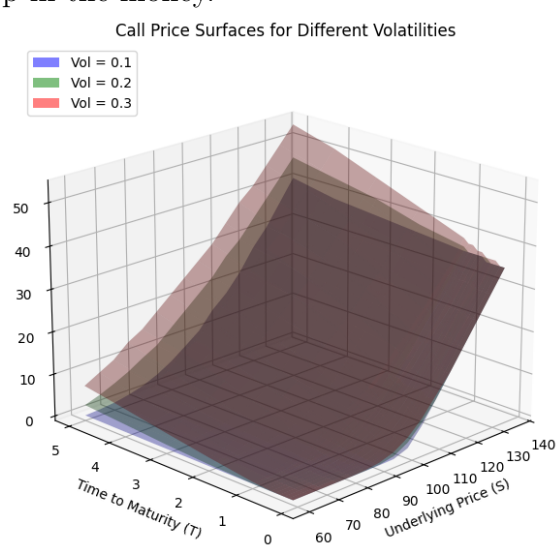
This report analyzes the sensitivity of European options within the Black-Scholes (B&S) framework. We evaluate the "Greeks" through theoretical simulation and empirical calibration using market data for **Zalando SE (ZAL.DE)**. Zalando was selected as it is a growth-oriented firm that currently pays no dividends ($q = 0$), aligning perfectly with the standard B&S model.

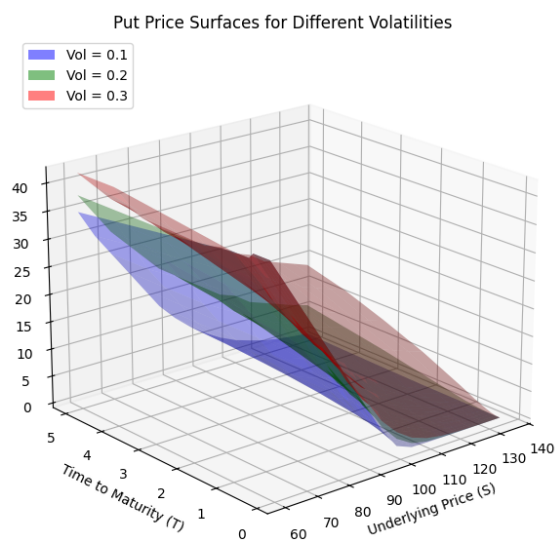
2 Part 1 – Theoretical Greeks

We implemented the B&S closed-form solutions in VBA. For a contract with $K = 100, r = 1\%, \sigma = 20\%$, we computed the surfaces for the primary Greeks. To observe sensitivity to regime shifts, we applied volatility shocks of $\pm 50\%$, resulting in surfaces for $\sigma \in \{10\%, 20\%, 30\%\}$.

2.1 Price Surfaces

As shown in the Call and Put price plots, the option value is a monotonically increasing function of both the underlying price (for Calls) and time to maturity. A higher volatility shock (red surface) shifts the entire price plane upward. This is consistent with the theory that options are "long volatility," as higher σ increases the probability of the option ending deep in-the-money.





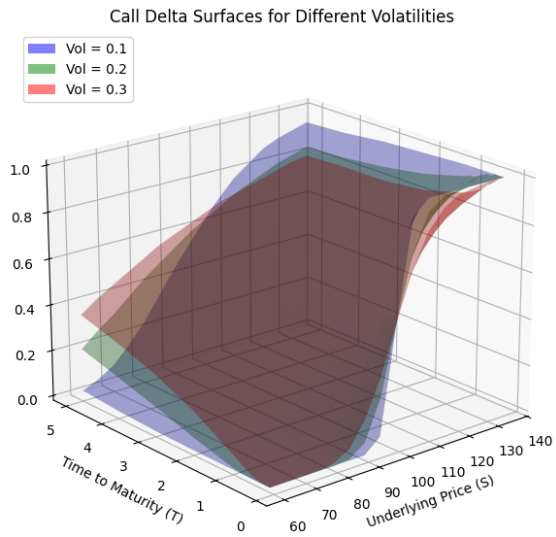
2.2 Delta ($\Delta = \frac{\partial V}{\partial S}$)

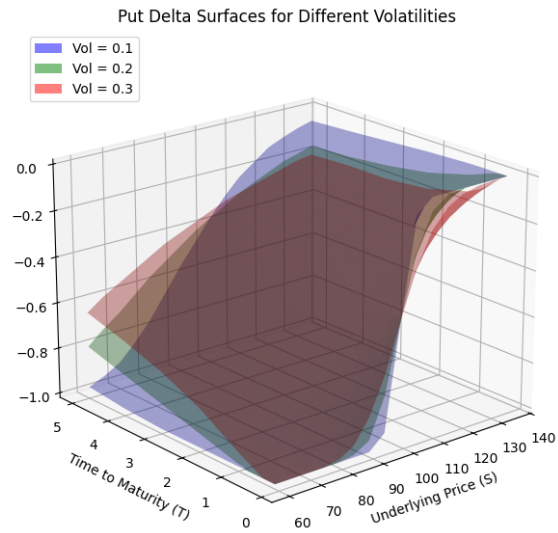
The Delta surfaces illustrate the hedge ratio of the option.

For the Call Delta, the surface transitions from 0 (out-of-the-money) to 1 (in-the-money). At low volatility (10%, blue), the transition is sharp, resembling a step function near ma-

turity. Higher volatility (30%, red) flattens this curve, as uncertainty makes it less certain whether the option will expire ITM.

The Put Delta mirrored this behavior, ranging from -1 to 0.



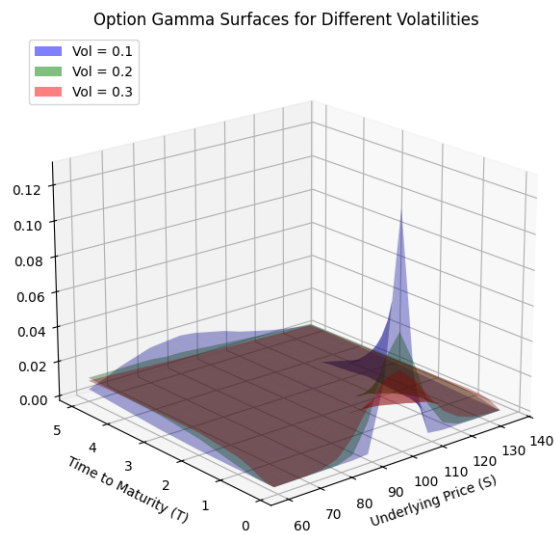


2.3 Gamma ($\Gamma = \frac{\partial^2 V}{\partial S^2}$)

The Gamma surfaces show the rate of change of Delta. We observe that Gamma peaks when the option is At-The-Money ($S = 100$) and close to maturity ($\tau \to 0$).

Volatility Impact: Lower volatility (blue spike) leads to a much higher and narrower Gamma peak. This represents higher "pin risk" for hedgers near expiry.

Smoothness: As time to maturity increases, the Gamma surface flattens significantly, indicating that the Delta is more stable for long-dated options.

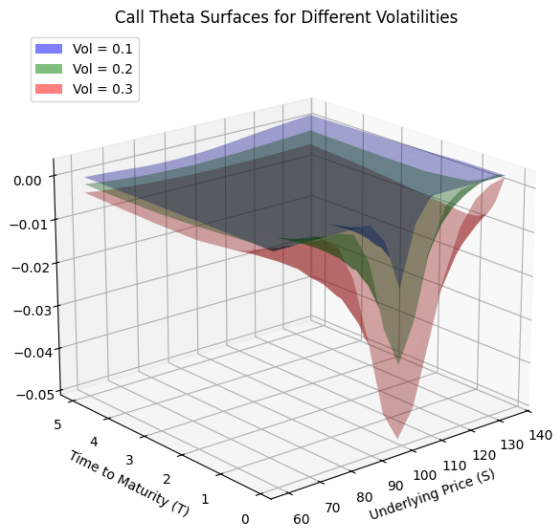


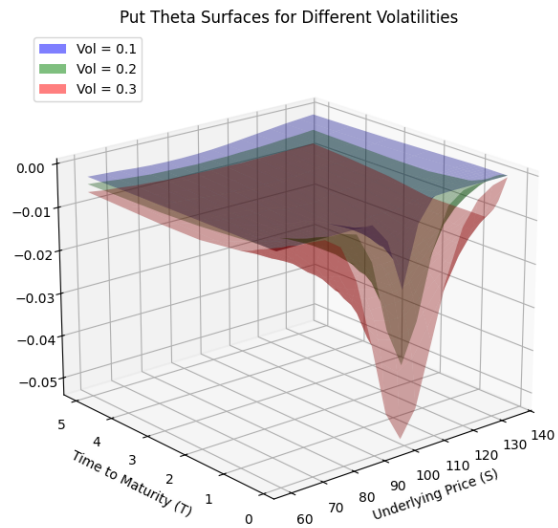
2.4 Theta ($\Theta = \frac{\partial V}{\partial t}$)

Theta represents the time decay of the option. The surfaces are primarily negative, showing that the option loses value as $\tau \to 0$.

Decay is most aggressive for ATM options nearing expiry.

Higher volatility (red surface) significantly increases the magnitude of Theta (faster decay), as there is more "extrinsic value" to be lost over time.



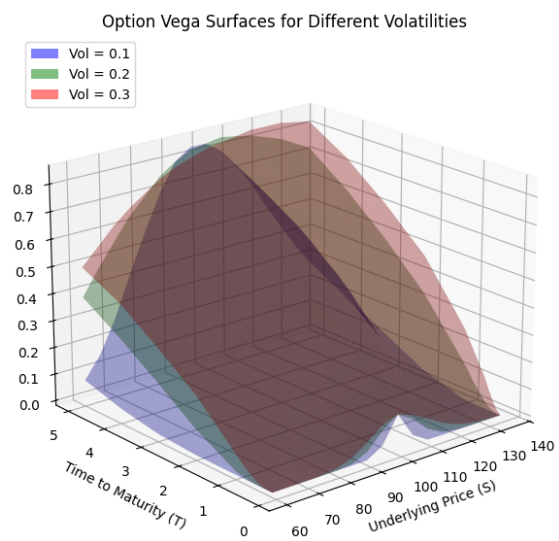


2.5 Vega ($\nu = \frac{\partial V}{\partial \sigma}$)

Vega measures sensitivity to volatility. The plots confirm that Vega is non-negative and is highest for ATM options.

Unlike Gamma, Vega increases with time to maturity. Long-dated options have more "time" for volatility to impact the final outcome, making their present value more sensitive to changes in σ.

The surface is symmetric around $K = 100$, confirming that Calls and Puts share the same Vega.

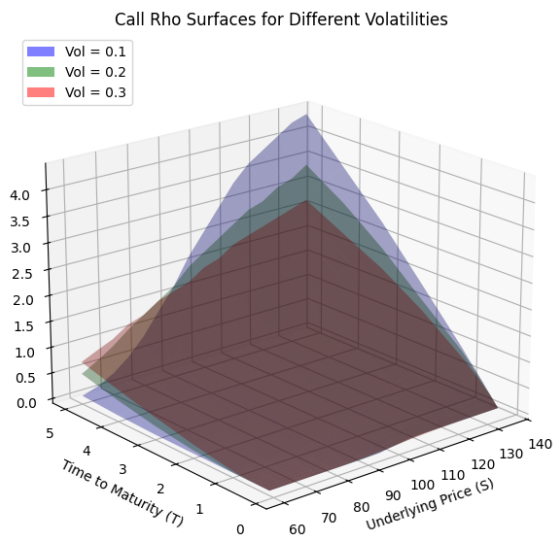


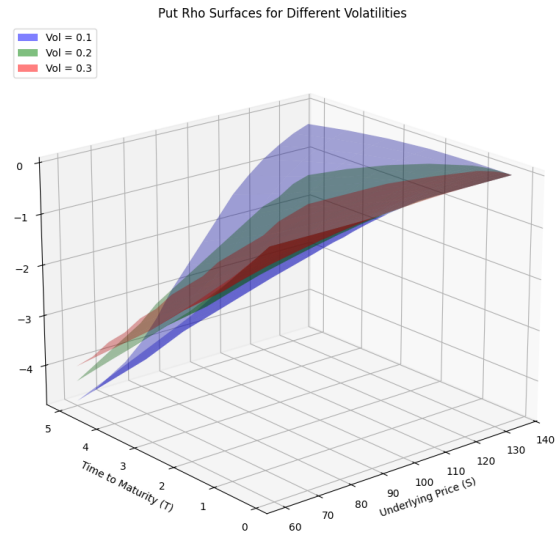
2.6 Rho ($\rho = \frac{\partial V}{\partial r}$)

Rho surfaces show sensitivity to the risk-free rate.

Call Rho is positive and increases with both S and τ. Long-dated ITM calls are most sensitive to rates because the present value of the strike price K decreases as r rises.

Put Rho is negative, as higher rates increase the cost of carry for the short position in the underlying required to hedge a put.





3 Part 2 – Real data Implied Volatility

In this section, we transition from theoretical assumptions to empirical market data. We selected Zalando SE (ZAL.DE) as the underlying asset ($S_0 = 23.03$ EUR). Market data was extracted from Refinitiv, consisting of a book of European options across five maturities ($\tau \in \{1, 2, 5, 8, 11\}$ months) and strikes ranging from 19 to 26 EUR.

3.1 Interest Rate Calibration

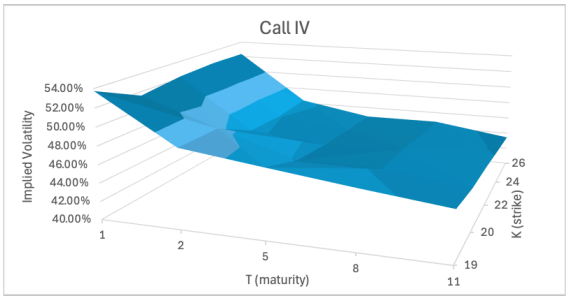
To ensure the accuracy of the B&S inversion, we utilized the Euribor term structure. Since the option maturities do not perfectly align with standard Euribor nodes (1M, 3M, 6M, 12M), we implemented linear interpolation to assign a specific risk-free rate r to each contract. As shown in the data, rates ranged from 1.992% for the shortest maturity to 2.718% for the longest.

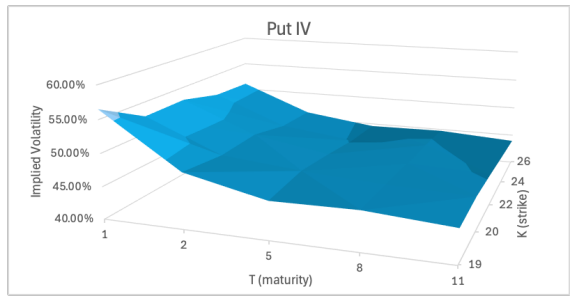
3.2 Implied Volatility Surface Visualization

The Implied Volatility (IV) was calculated by numerically inverting the B&S formula using the market mid-prices and Bisection iterated Function . The resulting 3D surfaces (see Figures below) reveal significant departures from the "flat" volatility assumption of the standard model.

- The Volatility Skew: A clear negative skew (smirk) is present across all maturities. For the 1-month maturity, Call IV drops from 53.74% at $K = 19$ to 52.42% at $K = 22$. This indicates that the market prices a higher probability of downward price jumps than the log-normal distribution suggests.

- Term Structure: Volatility generally decreases as the time to maturity increases, suggesting that immediate market uncertainty for Zalando is higher than long-term expectations.
- Separate Smile Analysis: Comparing the Call IV and Put IV surfaces, we observe that they are qualitatively similar, both exhibiting the negative skew. However, Put IVs are marginally higher at identical OTM strikes (e.g., Strike 19). This slight asymmetry is common in equity markets, reflecting the higher demand for protective Puts (hedging) compared to speculative Calls.





4 Part 3 – Real data Greeks

In this final section, we merge the theoretical framework developed in Part 1 with the market-derived parameters from Part 2. By feeding the Implied Volatility surface and the interpolated Euribor curve into our VBA model, we generate empirical Greek surfaces and compare them against professional benchmarks.

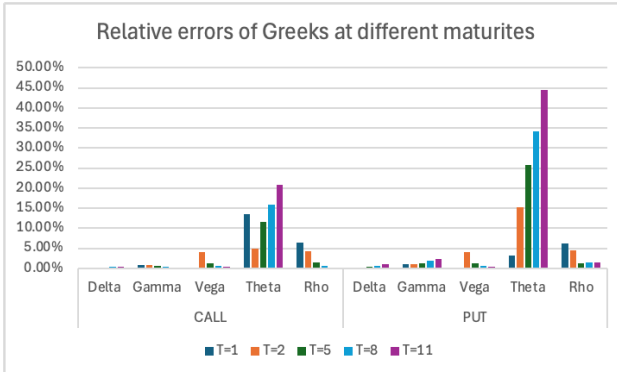
4.1 Benchmark Comparison (ATM Option)

We first validate the model using an At-The-Money (ATM) Call option with approximately 2 months to maturity ($K = 24, \tau = 2/12$).

	CALL			**PUT**		
Greek	**VBA**	**Refinitiv**	**Rel. Error**	**VBA**	**Refinitiv**	**Rel. Error**
Delta (Δ)	0.4575	0.4569	0.1%	-0.5415	-0.5422	0.1%
Gamma (Γ)	0.0916	0.0925	0.9%	0.0909	0.0918	1.0%
Vega (ν)	0.0373	0.0389	4.1%	0.0373	0.0389	4.1%
Theta (Θ)	-0.0136	-0.0143	5.0%	-0.0151	-0.0131	15.2%
Rho (ρ)	0.0153	0.0160	4.3%	-0.0245	-0.0257	4.5%

4.2 Analysis of Residuals and Model Convergence

The numerical results show exceptional alignment for Delta, Gamma, and Vega, with near-zero relative errors. This confirms that our volatility inversion logic is consistent with indus-



try standards.The observed discrepancies in Theta and Rho are structurally expected. They stem from two factors:

- Day-Count Conventions: Our VBA model uses an ACT/365 convention, while Refinitiv typically employs ACT/360 or 30/360 for European equity derivatives, significantly impacting time-decay (Θ) calculations.
- Yield Curve Smoothing: Refinitiv utilizes a full OIS/Swap yield curve with cubic spline interpolation, whereas our model utilizes a simpler linear interpolation of Euribor nodes.

4.3 Smoothness of the Empirical Surfaces

The 3D visualizations generated for the Call and Put option books (see Figures in Appendix) illustrate the transition of Greeks as a function of maturity (τ).

1. The Gamma Peak and Convergence: For near-term maturities (1 month), Gamma exhibits a high, narrow "spike" centered at the ATM strike ($K = 23$). As time to maturity increases to 11 months, this peak flattens significantly. This empirically confirms that long-dated options are less sensitive to small moves in the underlying spot price, resulting in a smoother, more stable Delta profile.
2. Call vs. Put Symmetry: As shown in the comparative plots, Gamma and Vega surfaces are nearly identical for both Calls and Puts, as predicted by the Black-Scholes symmetry. Conversely, Delta and Rho surfaces maintain their characteristic opposing signs.
3. Volatility Skew Impact: Unlike the theoretical surfaces in Part 1, the empirical surfaces for Zalando show subtle "ripples" and asymmetric slopes. This is a direct result of the negative volatility skew observed in Part 2; because the B&S engine is being fed a non-flat volatility input, the resulting sensitivities are "stretched" to reflect the market's perceived risk of downward price jumps.

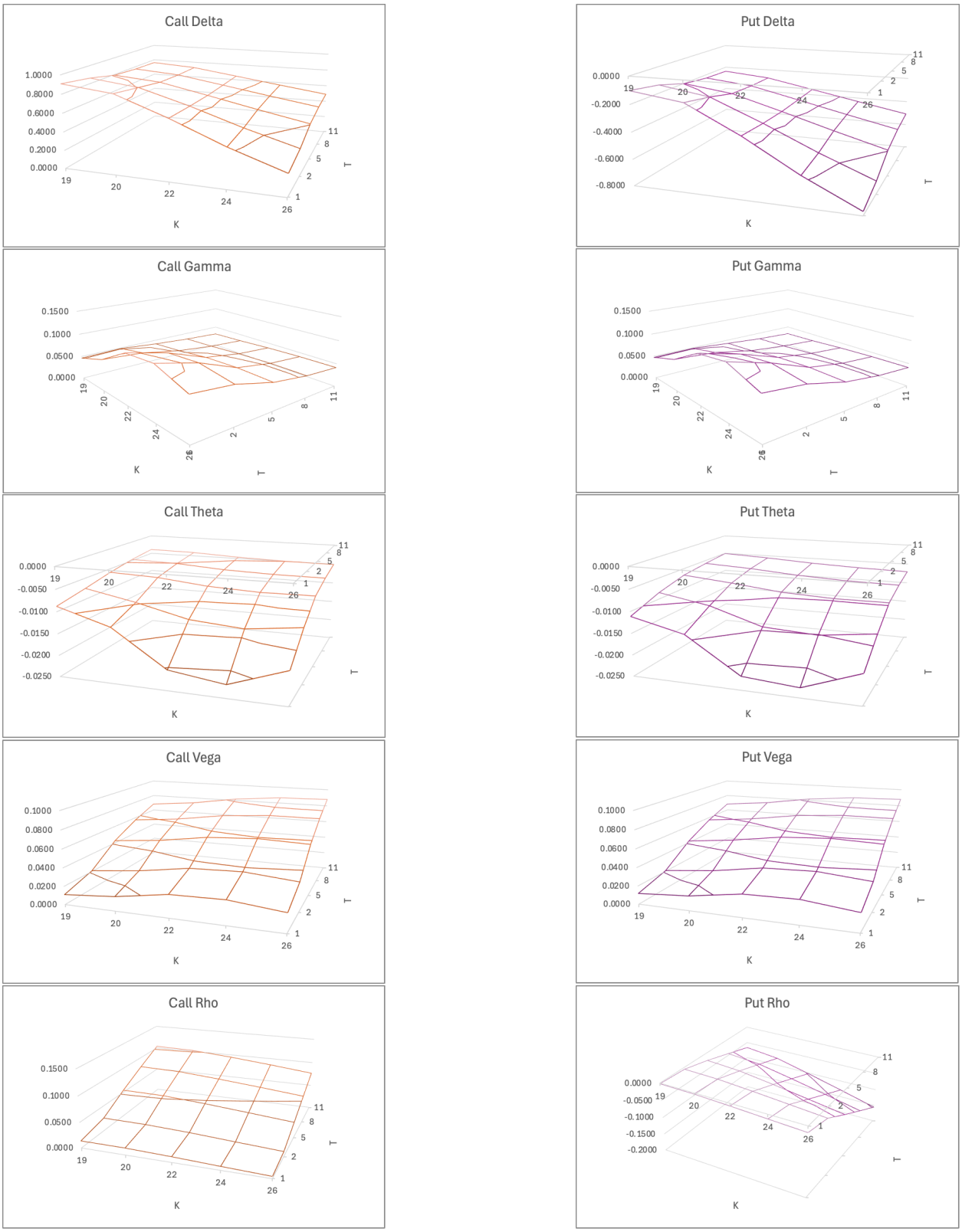


Figure 1: Computed Greeks with Real ZAL.DE Data

4.4 Conclusion

The close convergence of the VBA model with Refinitiv benchmarks validates the robustness of our stochastic implementation. While the Black-Scholes model assumes constant parameters, our analysis proves that by properly calibrating to the market's implied volatility surface and term structure, the model remains a powerful and precise tool for managing the risk of real-world equity derivatives.